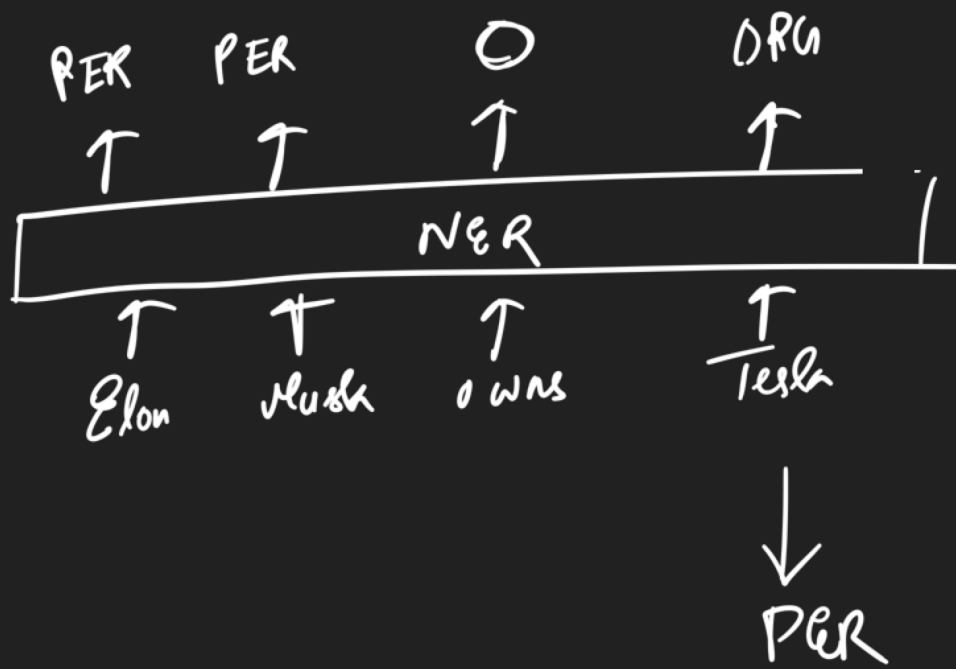




O : Other/Misc

Use case:

SYMPTOMS
MEDICINES
DOSAGES

} Named Entity
diffs

X	Y
✓ Elon is CEO	PER Other DESIGN ✓

✓ Seq classificatⁿ
→ Sentiment analysis
Many to One

Token class
→ NER

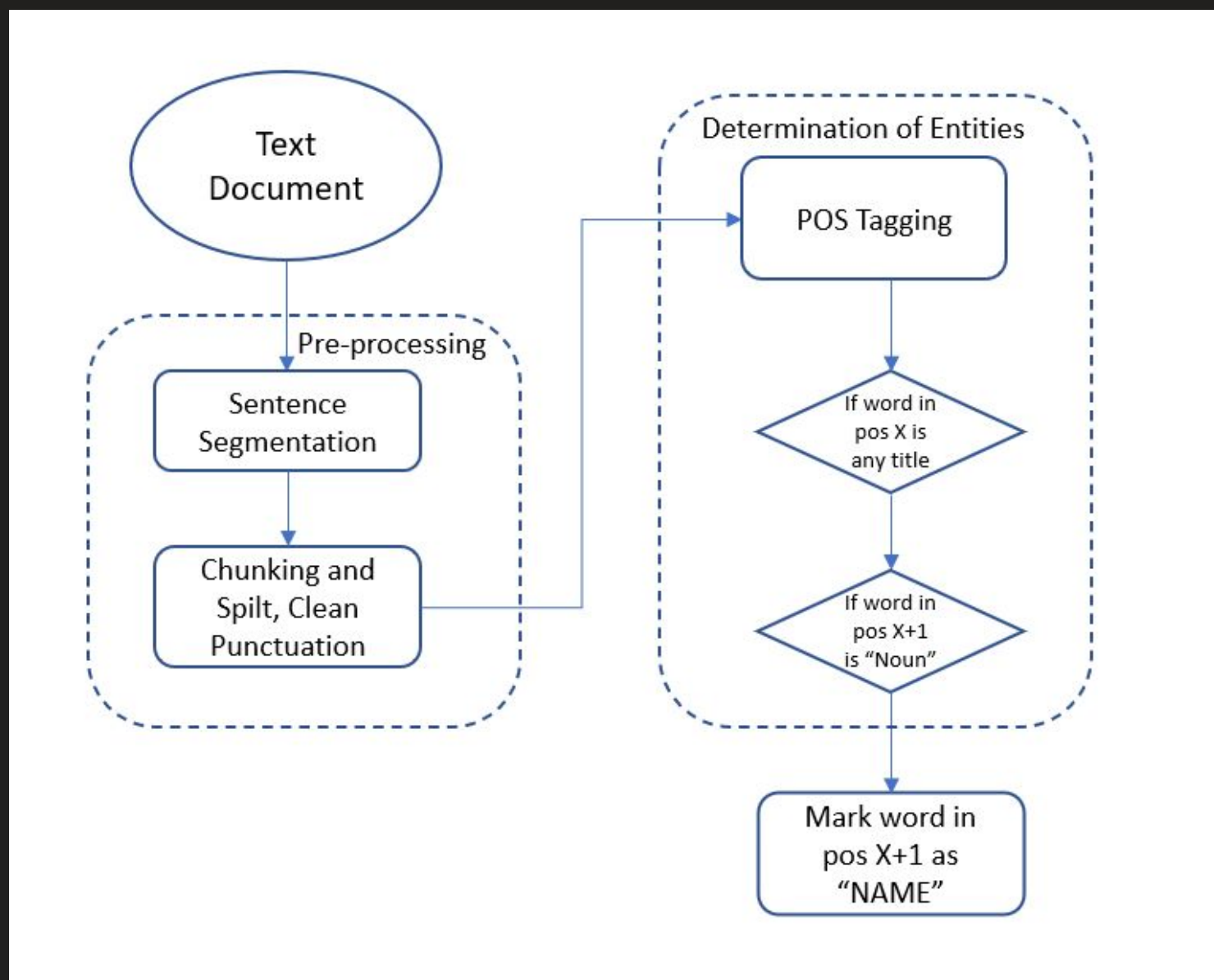
Many to Many
 $N_{in} = N_{op}$ } NER

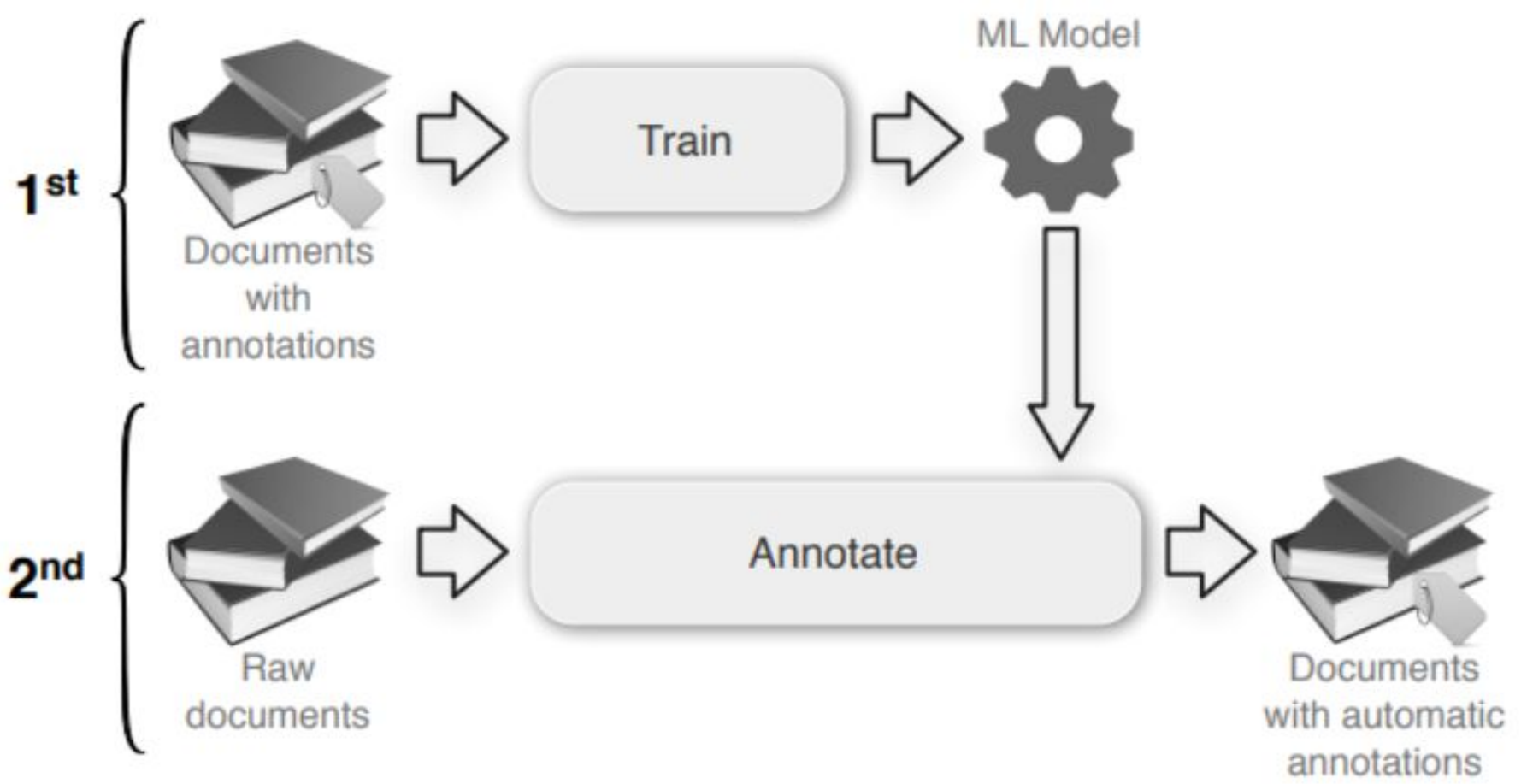
$N_{in} \neq N_{op}$ } translation

Pos tagging - NER - Token classificatⁿ algo

REF ↑ ↑ ↑ ↑ ↑ } POS
P The Sun is a God } +ve
↓ ↓ ↓
God God

Sentiment analysis → Sequence classificatⁿ
NER/POS → Token classificatⁿ





✓ Why is NER hard?

Some of the various ambiguities in NER include:

- **Segmentation(Detection) 'ambiguity'.**

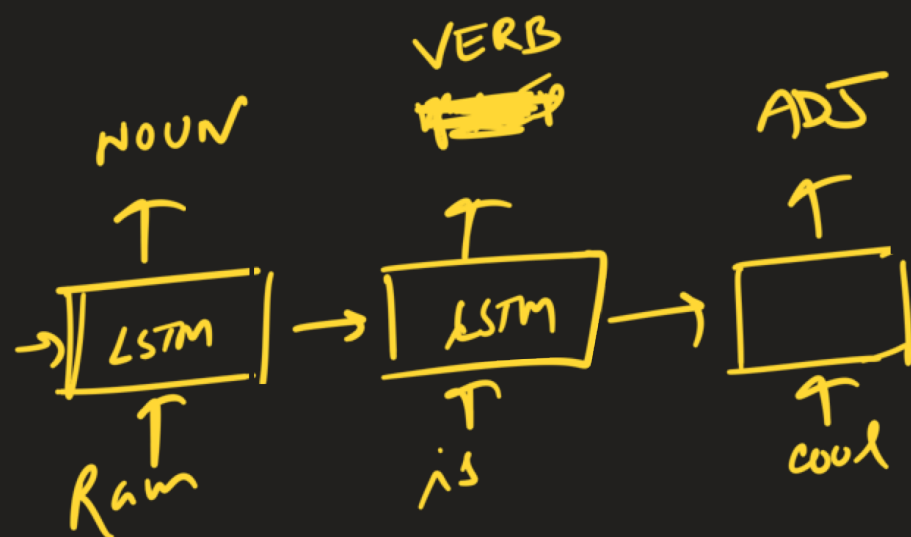
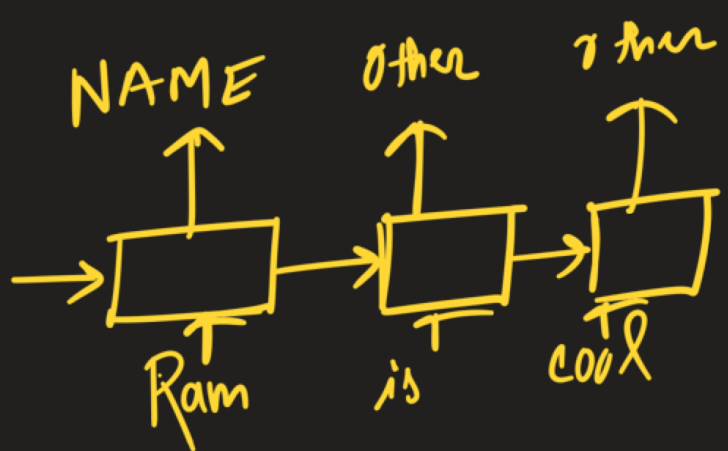
Consider 'New York'. This can be detected as 'New' & 'York' as separate entities.

Hence, deciding over the boundary is crucial whether to consider 'New York' as a single entity or 'New' and 'York' 2 different entities.

- **Tag assignment(Recognition) 'ambiguity'.**

We have 'Nirma' as a girl's name (PERSON)& as a detergent brand(Organization in India).

- In order to overcome these ambiguities, we introduce a new concept called **Sequence Labelling**



{ The
 a
 an } define

DEF → NOUN
 The sun rises in the East

LSTMS {

 → the word itself (Sun)

 → the label of previous word (DEF)

DEF NOUN
 The word

The Sun is a God

↓
DEFINITE

Model (Sun, DEFINITE) — NOUN

↑ label of prev word

↑ label of prev word

✓

w_t w_{t-1}
LSTM (Sun, The)

↑
prev word itself

only for
NER / POS ✓

Conditional Random Fields ✓

DEF → NOUN
↑ ↑
The Sun shines

LSTMS

DEF ~~X~~ NOUN
↑ ↑
The Sun shines

Why Conditional Random Fields?

- well suited for handling NER problems, because it takes **context into account**
- when a CRF model makes a prediction, it factors in the impact of neighbouring samples by modelling the prediction as a graphical model.
- A **linear chain CRF** confers to a labeller in which tag assignment(for present word, denoted as y_i) **depends only on the tag of just one previous word**(denoted by y_{i-1}).

Deciding Feature Functions:

We need to extract multiple features per word of the sentence that will be assisting in recognizing a Named Entity. For this, we need functions called as feature functions that will assist in generating unique features.

- These features function can consider any logic(depends on programmer) but the output has to be either True:1 or False:0.

$$P(LSTM_{w_t}) = f(w_t, w_{t-1})$$

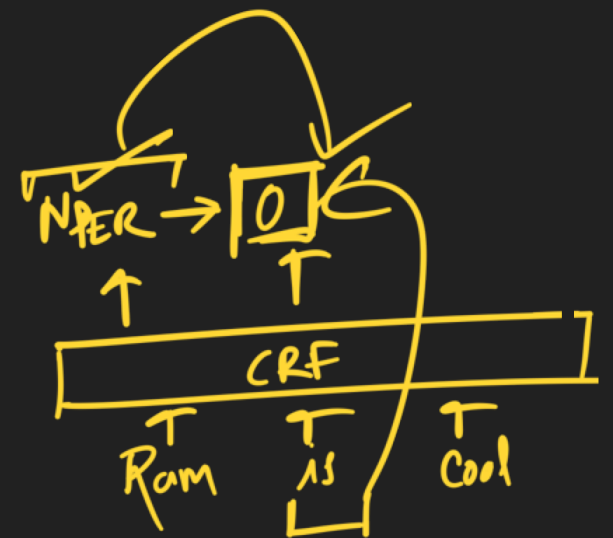
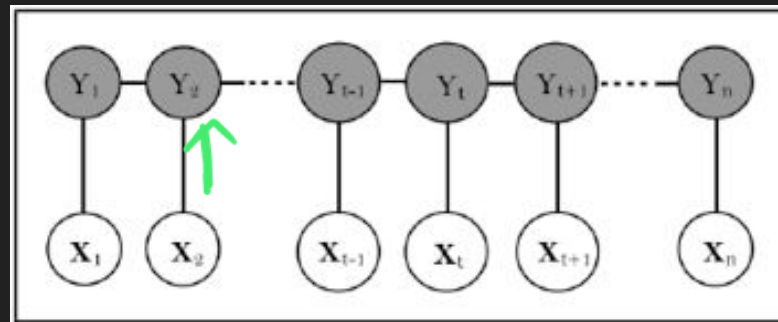
↓ label of previous word

$$P(NER_{w_t}) = f(w_t, y_{t-1})$$

↑ current word

↑
label of
current
word (w_t)

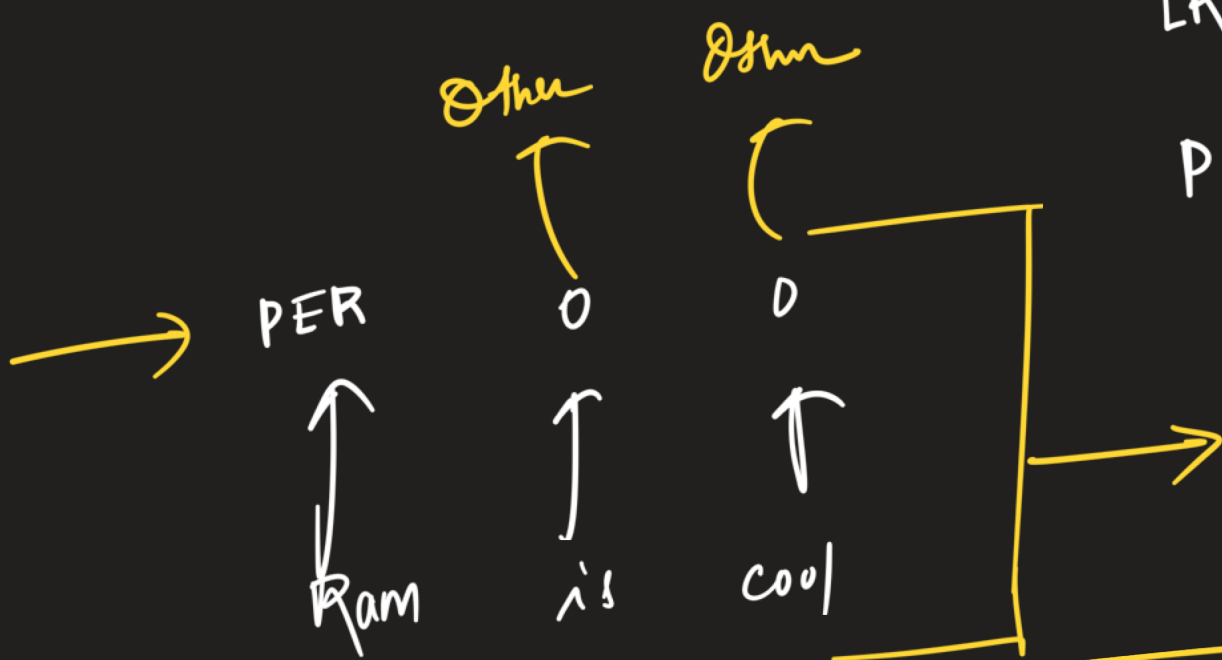
$y_1 \rightarrow y_2$



DEF → NOON
↑ ↑
The Sun shines

$$POS(sun) = f(sun, DEF)$$

LR $P(\text{Survival} | \text{features in titanic})$
 $P(\text{Defaulting} | \text{Salaries, Age})$



$P(\text{PER PER PER} | \text{Ram is Cool})$

$P(\text{PER } 0 \ 0 \text{Ram is cool})$

$P(\text{PER } 0 \ 0 | \text{Ram is cool})$ } maximum

seq of labels/
named entities text

of choices for the entire seq $\Rightarrow 2^{\boxed{3}} = 8$
 Ram is Col $\underline{\underline{S=3}}$

{ PER ✓
 O: Other ✓

PER	O	O
O	PER	O
PER	PER	O
⋮		
⋮		
⋮		

8 choices }

$$\frac{\exp(z_1)}{\exp(z_1) + \exp(z_2) + \exp(z_3)}$$



$z_1 \quad z_2 \quad z_3$

CRF $\rightarrow$ multiple feature functions — F_j F_1
 F_2

Consider the below formula:

CRF: $p(\text{PER } 0 \ 0 \mid \text{Ram is Cool})$

for all 2^n choices

$$p_{\theta}(y|x) = \frac{\exp(\sum_j w_j F_j(x, y))}{\sum_{y'} \exp(\sum_j w_j F_j(x, y'))}$$

, where $F_j(x, y) = \sum_{i=1}^L f_j(y_{i-1}, y_i, x, i)$

$F_j = \sum_{i=1}^L f_j$ } = [] vector or scalar

Here,

Ram is cool

$$F_j = \sum_{i=1}^3 f_j(w_i) = \text{isCap}(\text{Ram}) + \text{isCap}(\text{is}) + \text{isCap}(\text{Cool})$$

capital letter word

$$= 1 + 0 + 0$$

9:05

$p_{\theta}(y|x)$ refers to the probability of calculating a Label sequence(y) given a word sequence(x).

Text

2 ORGs

This is a partnership betn MS and OpenAg

✓
F1 -
F2 -
F3 -

Is the word in CAP = $\underline{1} + 0 + 0 + 0 + 0 + 1 + 0 + 1$
and after the word = $0 + 0 + 0 + 0 + 0 + 1 + 0 + 0$
and before the word = $0 + 0 + 0 + 0 + 0 + 0 + 0 + 1$

$$F_j(x, y) = \sum_{i=1}^L f_j(y_{i-1}, y_i, x, i)$$

F_1 is the word capitalized

$$\hookrightarrow F_1(\text{Ram is cool}) = 1 + 0 + 0$$

F_2
 $\uparrow$
 is the word preceded by Mn/Dn/Ms

$\text{Mn. } \text{Ebn} \text{ is a good CEO } \} = 0 + 1 + 0 + 0 + 0 + 0$

$$P(\text{"PER 0 0"} | \text{Ram is cool})$$

$$\exp \left(\sum_j w_j F_j \left(\text{"Ram is Cool"}, \text{PER 0 0} \right) \right)$$

$$= \frac{\exp \left(\sum_j w_j F_j \left(\text{"Ram is Cool"}, \text{PER PER PER} \right) \right)}{\exp \left(\sum_j w_j F_j \left(\text{"Ram is cool"}, \underbrace{0 \text{ PER PER}}_{\text{sequence change}} \right) \right) + \dots + 2^n}$$

$$P(\text{PER } 0 \ 0 \ 0 \mid \text{Ram is cool}) = \cancel{\sum} \exp(\text{Ram is cool} \mid 0 \ 0 \ 0) \cdot 0$$

$$= 0$$

$$\left\{ \begin{aligned} & w_1 F_1 (\text{Ram is cool} \mid \text{PER } 0 \ 0) + w_2 F_2 (\text{Ram is cool} \mid \text{PER } 0 \ 0) \\ & + \dots \end{aligned} \right. \\
 \hline
 \left\{ \begin{aligned} & \sum_{\substack{2^n \\ \text{augment}}} \sum_j w_j F_j (\text{Ram is cool} \mid \overbrace{\hspace{2cm}}^{\uparrow} \\ & \hspace{10cm} 2^3 \quad (\text{no. of serv}) \end{aligned} \right.$$

Automated feature extraction

$$F_j = f(w_1) + f(w_2) + f(w_3)$$

$$F_i = 1 + 0 + 0 \quad \text{Ram is cool}$$

(words being capital)

$$F_{\text{embeddings}} = \text{Word2vec}(\text{Ram}) + \text{Word2vec}(\text{is}) + \text{Word2vec}(\text{cool})$$

$$F_{\text{LSTM}} = \text{LSTM}(\text{Ram}) + \text{LSTM}(\text{is}) + \text{LSTM}(\text{cool})$$

NER — Character level info are important

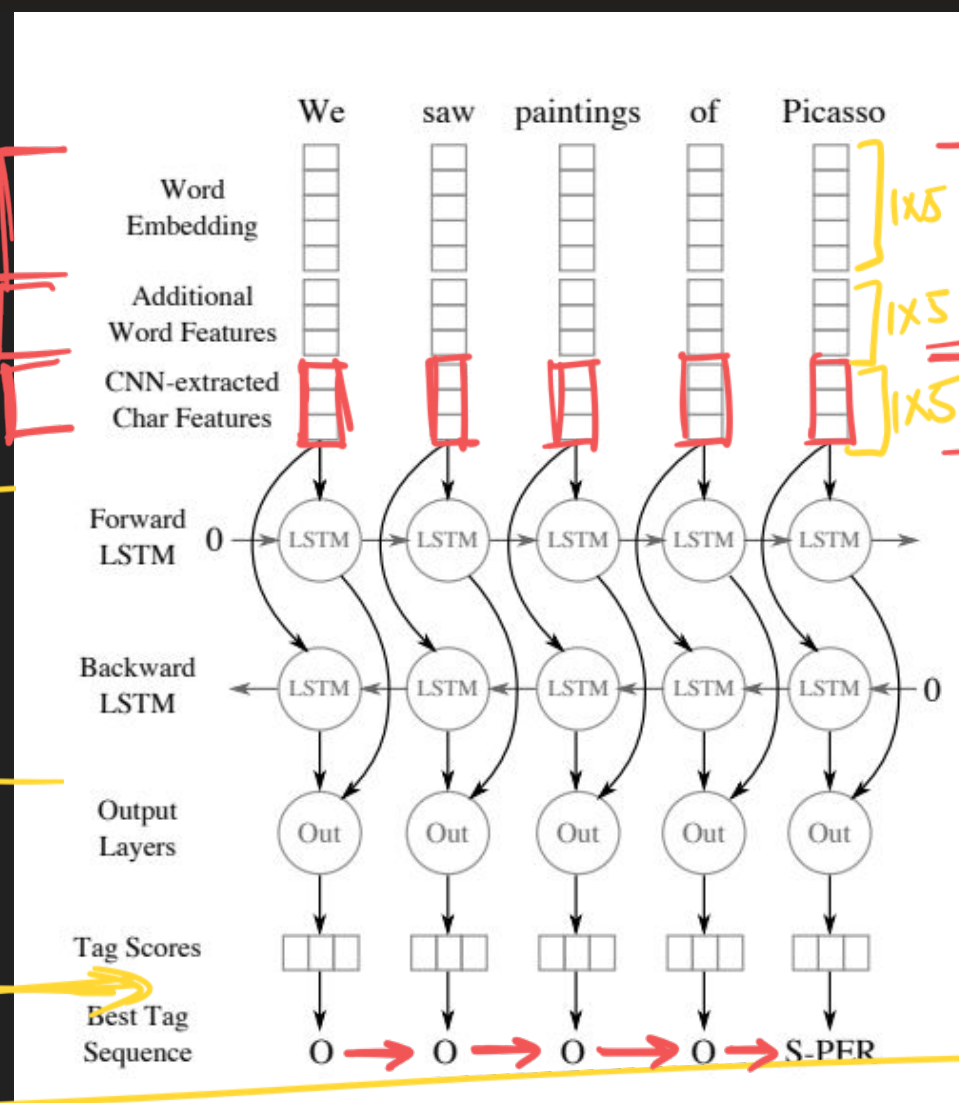
BERT — cased ✓

Sentiment Analysis — BERT — uncased

Wordvec + Glove

Bi-LSTM

no. of entity types



5x1

Word

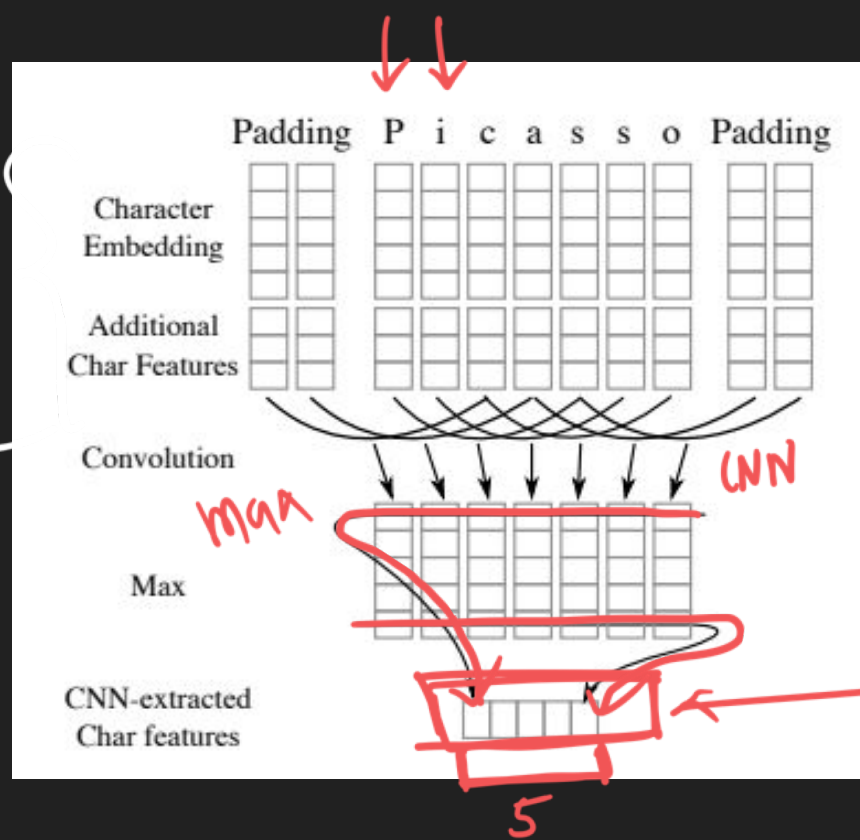
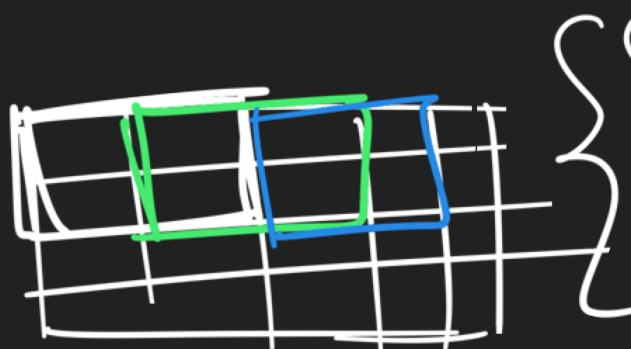
Character

1x1

5 outputs

3 logits [softmax]

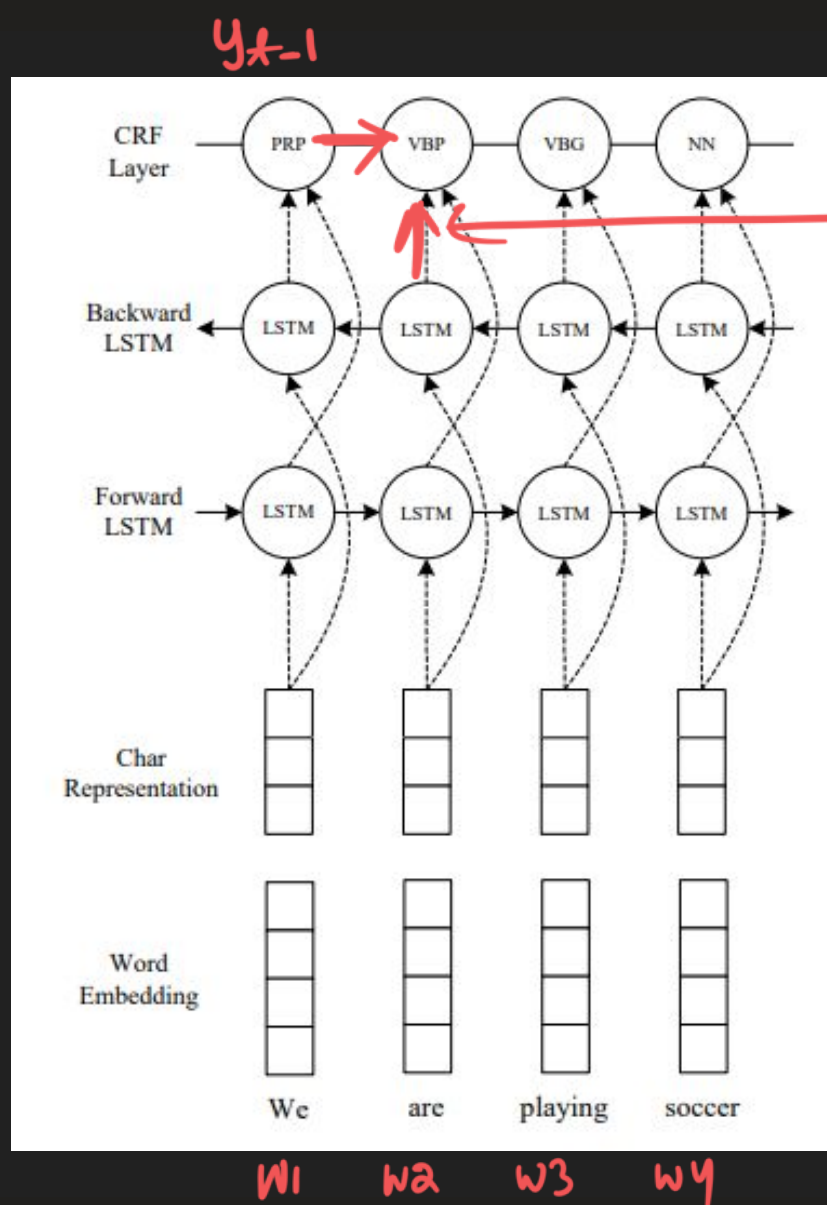
no. of numbers?



Open AG
↑↑
info

encoded character level info

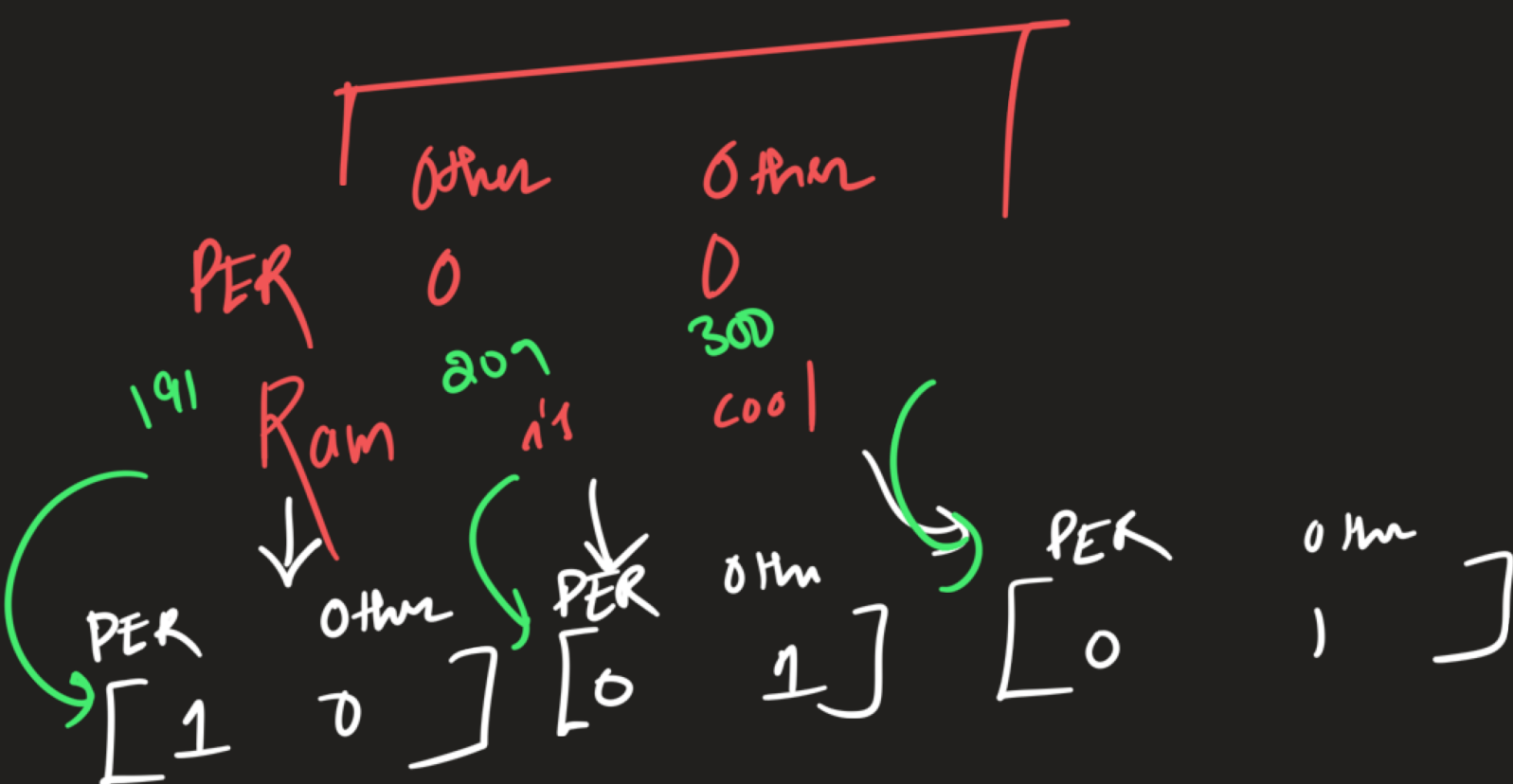
$$F(w_t, y_{t-1})$$



learned feature of w_t

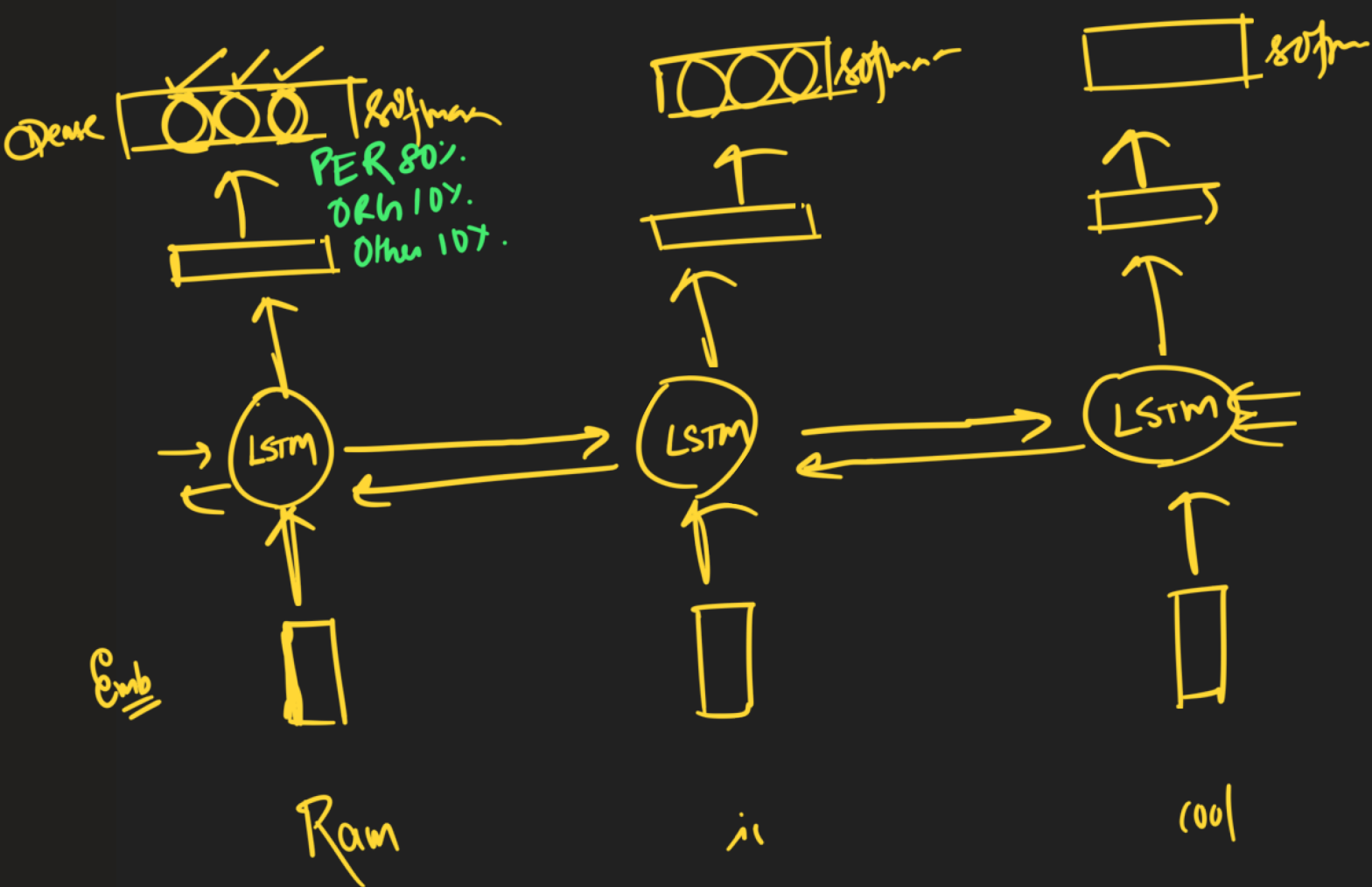
Bi-LSTM

Emb



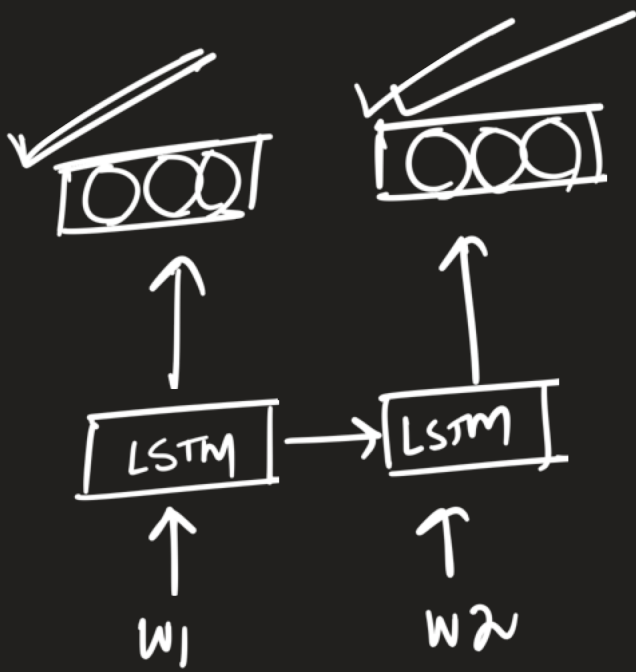
10 words - 2 NE + 8 Other

PER
0



ORG
PER
Other

no. of
NE



NEs

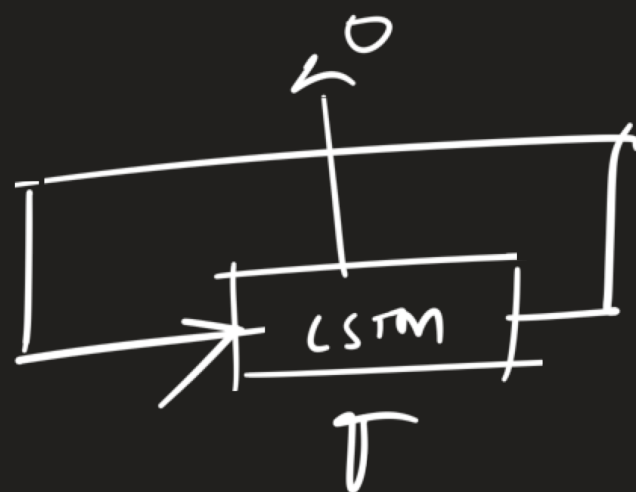
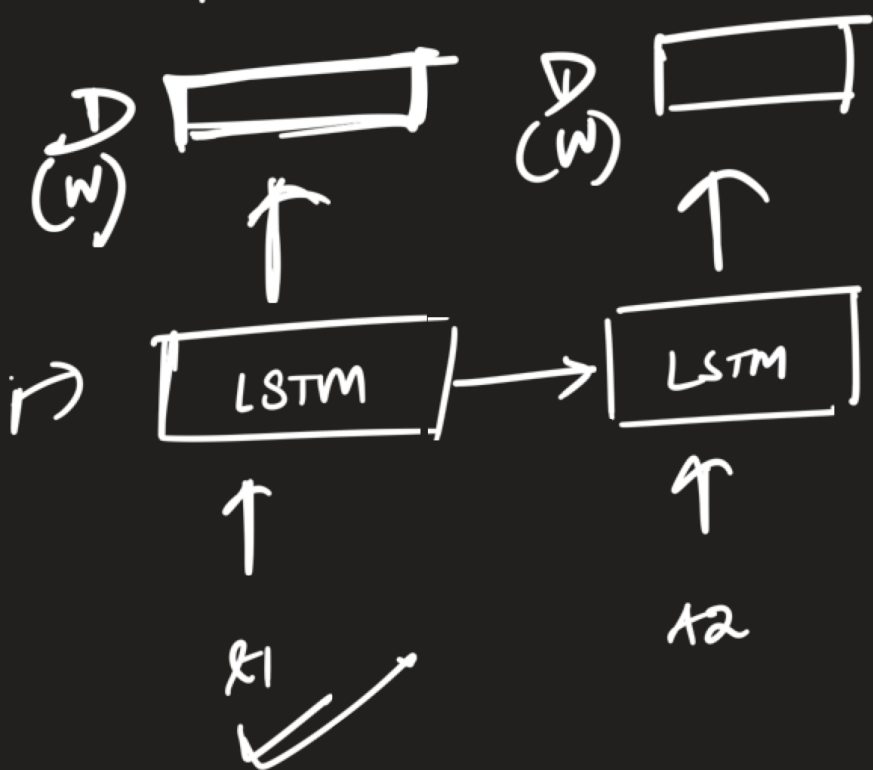
for n loop

same
diff



Overparameterization

Time Distributed



Same Dense layer
diff Dense lay

no diff time
acc diff time step

Next wed

Wed
Mon

NLP

{AS Eng - next Monday}

~~Transformers~~
↑

ATTN
Transformer ✓